

MAY 2026

PROJECT REPORT

CUSTOMER CONVERSION & CAMPAIGN FUNNEL ANALYSIS REPORT

Internship
Data Science and Analytics
Organization
Future Interns

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SUMMARY

This project focuses on analyzing customer behavior and campaign performance using the Bank Marketing dataset. The objective is to understand how customers respond to marketing efforts and identify key factors influencing conversion. The analysis was performed using Python (Google Colab) and Microsoft Excel, where data was cleaned, explored, and visualized through multiple charts and a funnel-based dashboard. The project highlights customer segments, campaign effectiveness, and conversion patterns to support better marketing decisions.

PROBLEM STATEMENT

Businesses often struggle to achieve high conversion rates from marketing campaigns due to a lack of clear insights into customer behavior. This leads to:

- Low campaign effectiveness
- Poor targeting of customers
- Inefficient use of marketing resources

This project aims to analyze campaign data to identify conversion patterns and improve marketing strategies.

PROJECT OBJECTIVES

- Analyze customer conversion behavior
- Identify key factors affecting campaign success
- Evaluate performance across contact methods
- Study customer segments and their response
- Build an interactive funnel dashboard
- Provide actionable business recommendations

DATA DESCRIPTION

The dataset used is the Bank Marketing dataset, which contains customer and campaign-related information.

It consists of approximately 4,500+ records with multiple attributes related to demographics, financial status, and marketing interactions.

Key attributes include:

- age – Customer age
- job – Profession
- marital – Marital status
- education – Education level
- balance – Account balance
- contact – Communication type (cellular, telephone)
- month – Last contact month
- campaign – Number of contacts performed
- poutcome – Outcome of previous campaign
- y – Target variable (Yes/No conversion)

DATA PREPROCESSING

Tools & Technologies Used:

- Google Colab (Python – Pandas, Matplotlib, Seaborn)
- Microsoft Excel (Pivot Tables, Charts, Slicers)

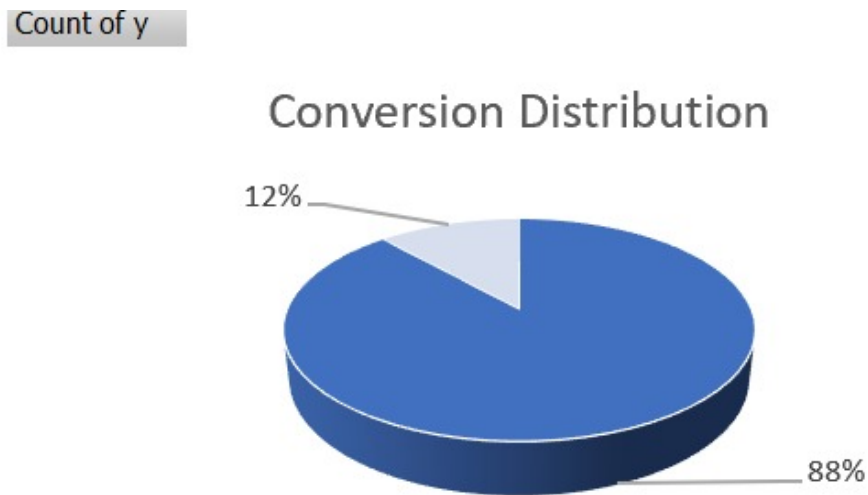
Steps Performed:

- Loaded dataset and fixed delimiter issues (semicolon separated)
- Split columns correctly for structured analysis
- Checked and handled missing values
- Verified data types and ensured consistency
- Created new feature:
- Balance Group (Negative, Low, High)
- Structured dataset for pivot analysis and dashboard creation

Data preprocessing ensured that the dataset was clean, well-structured, and ready for analysis and visualization.

EXPLORATORY DATA ANALYSIS (EDA)

1. Conversion Distribution



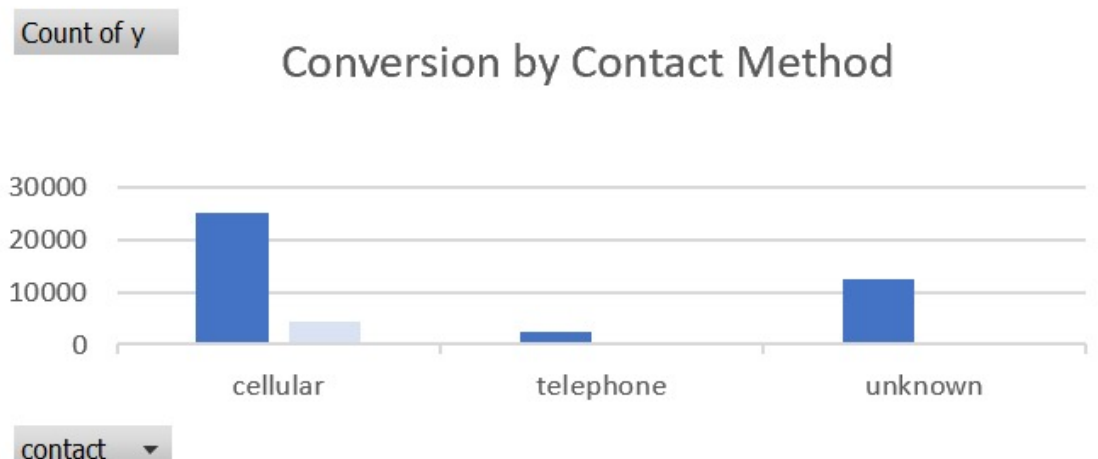
This chart shows the proportion of customers who subscribed versus those who did not.

Insights:

- Majority of customers did not subscribe
- Conversion rate is relatively low
- Indicates need for better campaign strategies
- Large gap between contacted and converted customers
- Highlights inefficiency in current marketing approach

EXPLORATORY DATA ANALYSIS (EDA)

2. Conversion by Contact Method



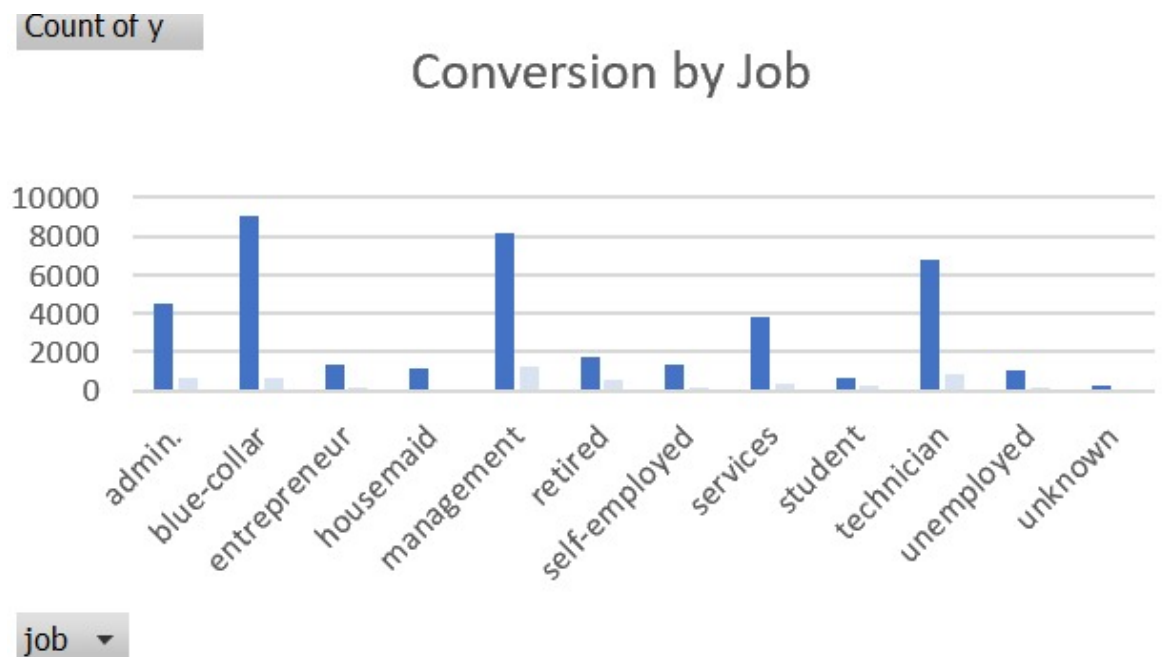
This chart compares conversion performance across different communication methods.

Insights:

- Cellular contact shows higher conversion rates
- Telephone contact performs moderately
- Unknown contact methods show poor results
- Communication method significantly impacts campaign success
- Direct and modern communication channels perform better

EXPLORATORY DATA ANALYSIS (EDA)

3. Conversion by Job



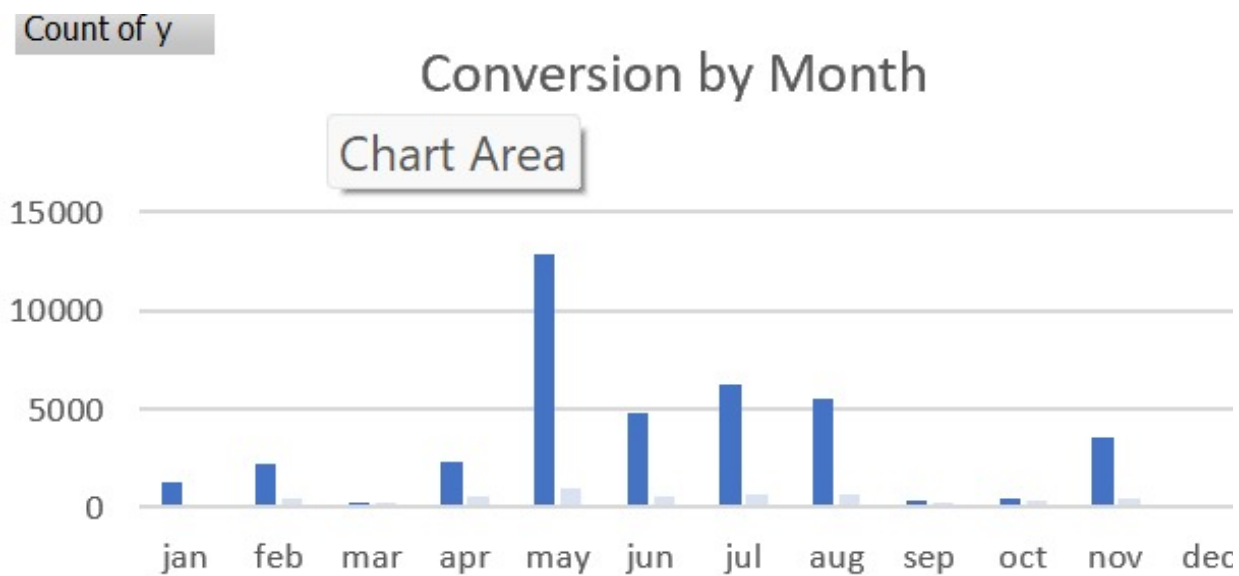
This chart analyzes how different job categories respond to campaigns.

Insights:

- Certain job groups show higher conversion rates
- Some segments contribute majority of conversions
- Helps identify high-value customer segments
- Job-based segmentation improves targeting
- Not all professions respond equally to campaigns

EXPLORATORY DATA ANALYSIS (EDA)

4. Conversion by Month



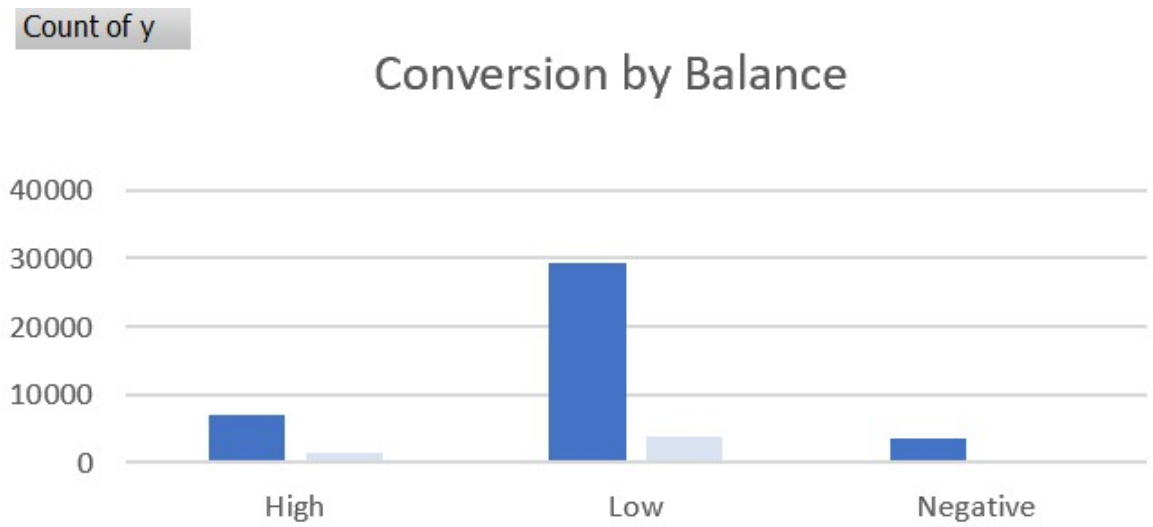
This chart shows conversion trends across different months.

Insights:

- Some months show higher conversion rates
- Campaign timing affects performance
- Seasonal trends may influence customer decisions
- Identifies best time to launch campaigns
- Poor-performing months indicate ineffective outreach timing

EXPLORATORY DATA ANALYSIS (EDA)

5. Conversion by Balance

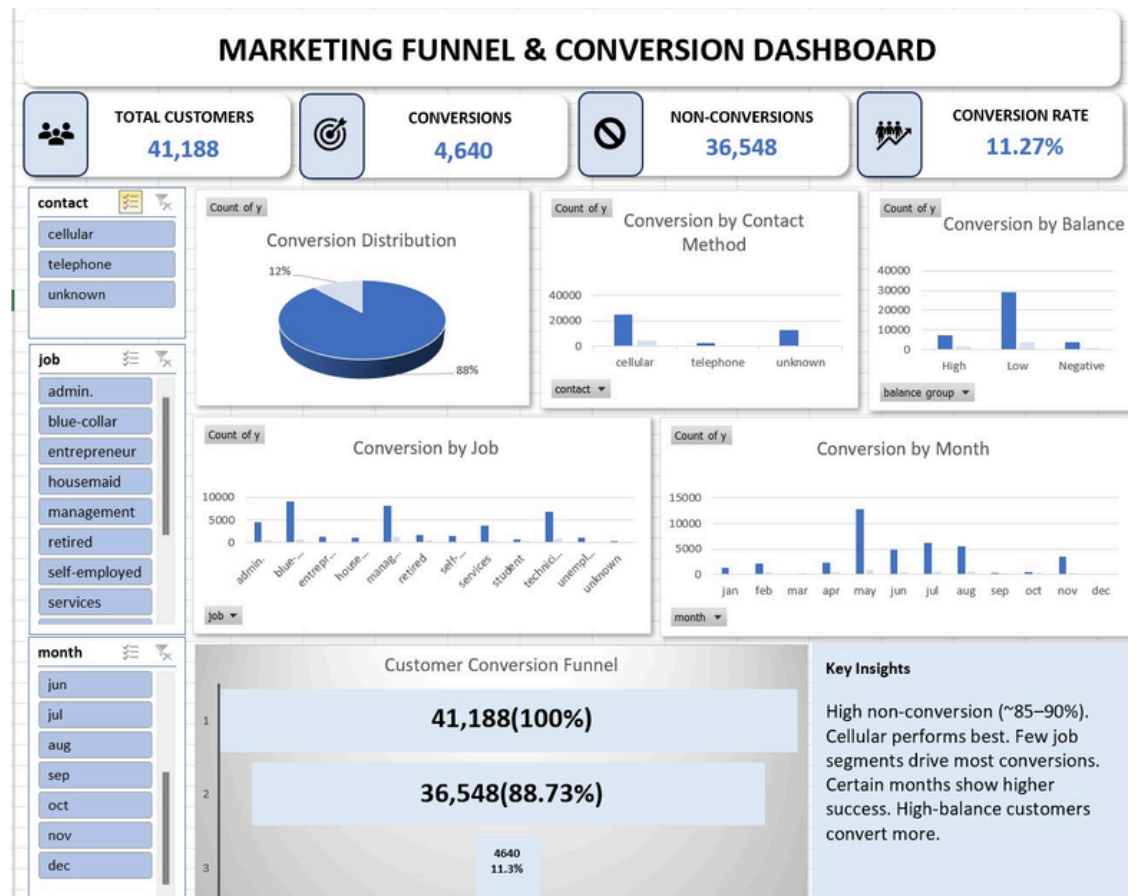


This chart analyzes how customer financial status affects conversion.

Insights:

- Customers with higher balance convert more
- Low or negative balance customers convert less
- Financial stability influences decision-making
- High-value customers are more likely to respond
- Balance is a strong indicator of conversion potential

DASHBOARD



The dashboard presents a comprehensive view of campaign performance and customer conversion behavior through multiple visualizations. It includes charts for Conversion Distribution, Conversion by Contact Method, Conversion by Job, Conversion by Month, and Conversion by Balance, along with a funnel chart to illustrate the drop from total contacted customers to converted customers. Interactive slicers such as Job, Contact, and Month allow users to dynamically filter data and explore different segments. The dashboard is designed with a clean and structured layout, making it easy to identify patterns, compare performance, and derive insights. It serves as an effective decision-making tool for improving marketing strategies and increasing conversion rates.

KEY INSIGHTS

- High non-conversion rate (~85–90%)
- Cellular contact performs better than other methods
- Few job segments drive majority of conversions
- Certain months show higher campaign success
- High-balance customers are more likely to convert

BUSINESS RECOMMENDATIONS

- Focus on high-performing contact methods like cellular
- Target high-value customer segments based on job and balance
- Optimize campaign timing based on best-performing months
- Improve strategies for low-performing segments
- Personalize campaigns to increase engagement and conversion

CONCLUSION

This project demonstrates how funnel and campaign analysis can help businesses understand customer behavior and improve marketing effectiveness. By analyzing conversion patterns across different segments and factors, valuable insights were generated. The dashboard provides a clear visualization of customer journey and campaign performance, enabling data-driven decisions to enhance conversion rates and optimize marketing strategies.

THANKYOU